



EPC

AI-Powered Project Intelligence Layer for EPC Engineering

Google Gemini • Clause-Tree Structural Retrieval • Quality Compliance Automation

AegisEPC Platform

<https://github.com/Jaswanth258/AI-Data-Centre-EPC-Platform>

THE CONTEXT & PROBLEM

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67%

Overrun rate on large EPC
infrastructure projects

**3-5
Days**

Manual review time per
vendor submittal package

\$15B+

National capital at risk
from schedule delays

Engineers spend days manually cross-referencing submittals against technical specs. A single missed non-conformance (e.g. 8-min battery runtime vs 10-min spec) creates severe delays and rework when caught post-delivery.

 RFI & Knowledge Copilot

Natural language interface querying technical guidelines with exact clause-by-clause traceability.

- Interactive citation chips linking to full spec views
- Per-document isolated chat history
- Dynamic domain adaptation based on loaded metadata

 Quality Compliance Agent

Automatic verification of vendor specs against ground-truth technical parameters.

- Fast, automated param extraction
- Clause comparison with deviation detection
- Status verdicts and severity metrics with CSV exports

✗ Traditional Vector RAG

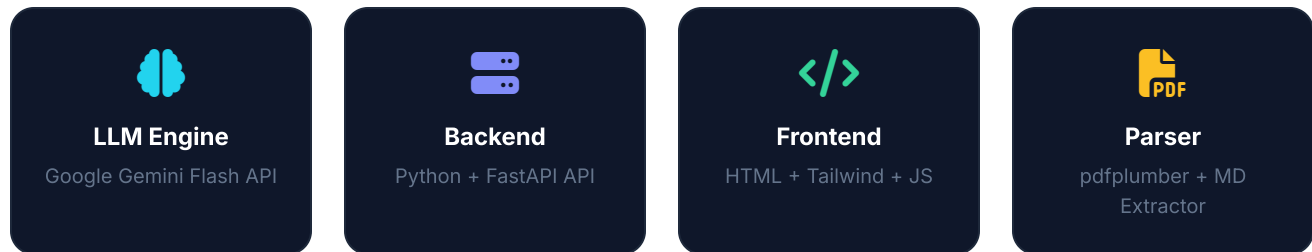
- Documents split into arbitrary 500-token chunks
- Chunks cut sentences/lists halfway, destroying context
- Cosine similarity loses semantic relationships
- Hard to verify citations back to exact legal sections

✓ AegisEPC Clause-Tree RAG

- Parser maps headers to a nested clause tree
- Preserves document hierarchy (Section → Clause X.Y)
- LLM navigates TOC, then extracts target clause text
- 100% deterministic, zero embeddings, zero vector DB cost

TECH STACK & PIPELINE

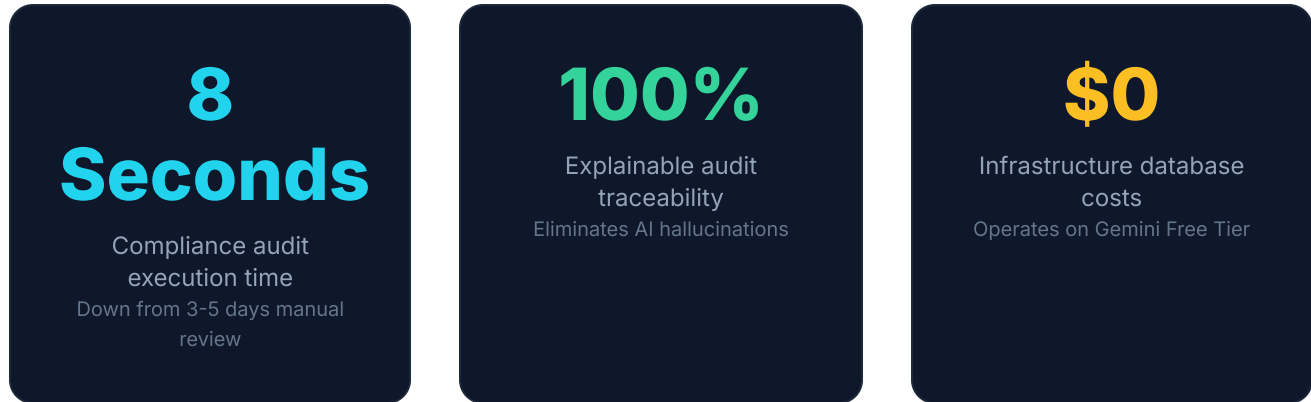
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The compliance agent pipeline uses a multi-agent workflow: Step 1 extracts params from submittals → Step 2 navigates the clause tree → Step 3 parses matching specification clauses → Step 4 verifies compliance status and outputs structure.

BUSINESS IMPACT

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By accelerating standard engineering review cycles from days to seconds, AegisEPC removes bottleneck delays from procurement pipelines, preventing project schedule overruns and ensuring zero cost impact from post-delivery equipment rejections.

THANK YOU

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AegisEPC Platform Presentation

GitHub Repo: [Jaswanth258/AI-Data-Centre-EPC-Platform](https://github.com/Jaswanth258/AI-Data-Centre-EPC-Platform)

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